



MARS! / SCIENTIFIC MISSION

MISSION No.

13

RED DUST CITY

ENERGY BUDGET

CENTRAL QUESTION

How much of our lives runs on electricity, and what is indispensable for us?

MORNING

ENERGY – FOOTPRINT – ENERGY BALANCE

- Read the power sockets / meters (record the values).
- Document every device.
- Categorize devices by function: eating / living / research / body / communication.
- Calculate how many Wh an activity costs:
 - a pot of coffee, one day of hydroponics, 1× feeding the grasshoppers, 1× eating.
- What does a Habitat need in order to live?

AFTERNOON

THE VALUE OF A KILOWATT-HOUR – decision

experiment
Scenario: We have 1 kWh per day. What can we continue doing, what should we stop?
→ coffee, hydroponics, hot plate, grasshoppers?

Create an individual plan and then make a joint decision.

HUMAN POWER – how much work is in a coffee?

- water – bicycle dynamo / exercise bike
- define amount & starting temperature
- 1 person
- time how long until the water boils
- repeat, change person
- How much human work is in everyday things?
- Which of our Earth habits would we want to pay for with energy?

EVA

ENERGY HUNT

Find:

- human energy (work, movement)
 - mechanical energy (bicycle, wheelchair, vehicles)
 - thermal energy
 - where energy is used in public space / everyday life
- 10 “Energy Encounters”

Ask: What would happen if tomorrow no electricity were available? What could be replaced with human labor, and what could not?

QUESTION FOR THE COMMUNITY HOUR

What do you use electricity for that you would not want to give up?

Welchen Prozess / welche Maschine, die auf Strom läuft, würdest du auf gar keinen Fall aufgeben wollen?

MATERIAL – WHAT DOES THIS NEED

Pen/Paper, laptop, electricity meters, power bike, pedometer
(?)